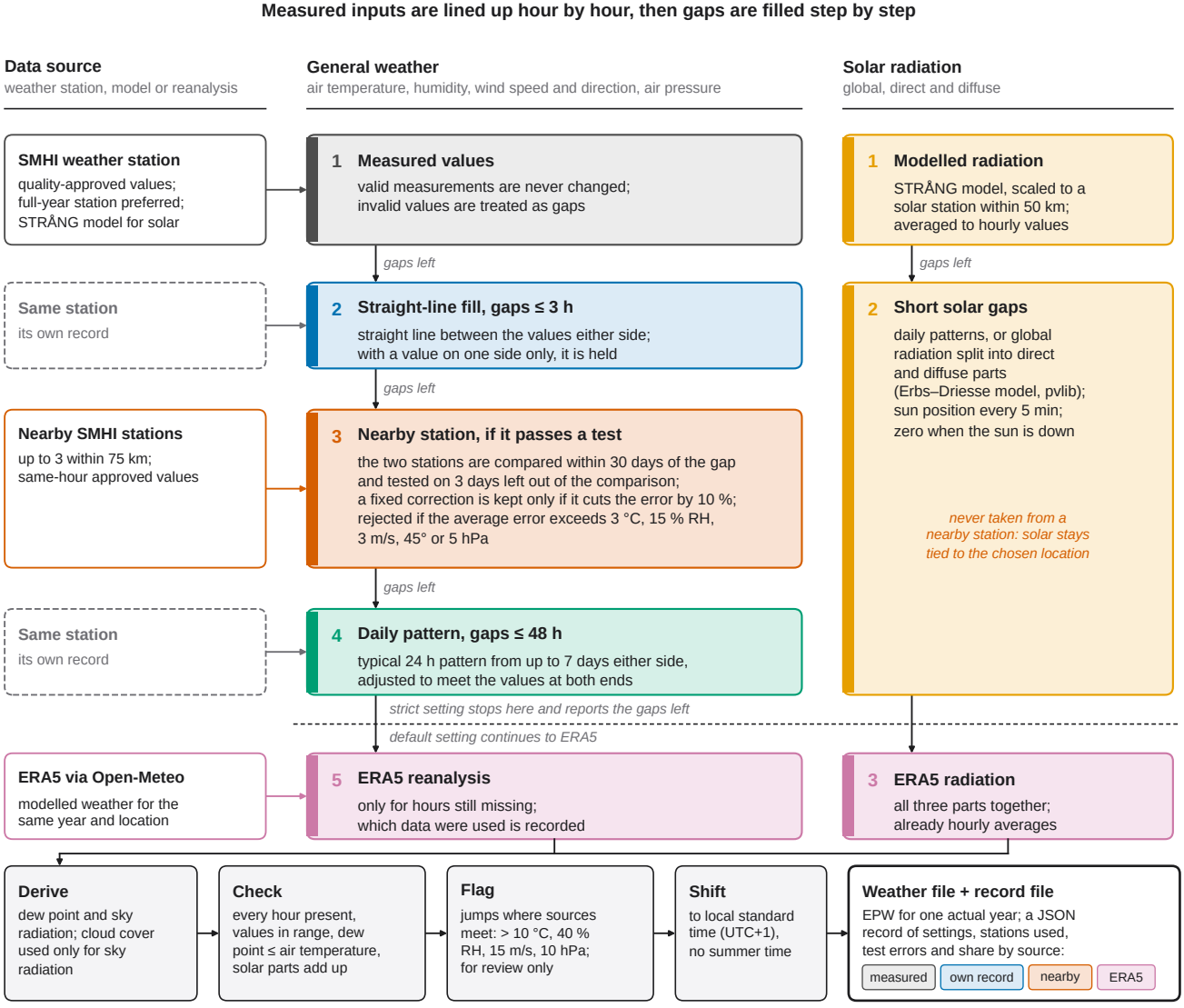
Measured inputs are lined up hour by hour, then gaps are filled step by step
Data source
weather station, model or reanalysis
General weather
air temperature, humidity, wind speed and direction, air pressure
Solar radiation
global, direct and diffuse
SMHI weather station
quality-approved values;
full-year station preferred;
STRÅNG model for solar
1 Measured values
valid measurements are never changed;
invalid values are treated as gaps
1 Modelled radiation
STRÅNG model, scaled to a
solar station within 50 km;
averaged to hourly values
gaps left
gaps left
Same station
its own record
2 Straight-line fill, gaps ≤ 3 h
straight line between the values either side;
with a value on one side only, it is held
2 Short solar gaps
daily patterns, or global
radiation split into direct
and diffuse parts
(Erbs–Driesse model, pvlib);
sun position every 5 min;
zero when the sun is down
never taken from a
nearby station: solar stays
tied to the chosen location
gaps left
Nearby SMHI stations
up to 3 within 75 km;
same-hour approved values
3 Nearby station, if it passes a test
the two stations are compared within 30 days of the gap
and tested on 3 days left out of the comparison;
a fixed correction is kept only if it cuts the error by 10 %;
rejected if the average error exceeds 3 °C, 15 % RH,
3 m/s, 45° or 5 hPa
gaps left
Same station
its own record
4 Daily pattern, gaps ≤ 48 h
typical 24 h pattern from up to 7 days either side,
adjusted to meet the values at both ends
strict setting stops here and reports the gaps left
default setting continues to ERA5
ERA5 via Open-Meteo
modelled weather for the
same year and location
5 ERA5 reanalysis
only for hours still missing;
which data were used is recorded
3 ERA5 radiation
all three parts together;
already hourly averages
Derive
dew point and sky
radiation; cloud cover
used only for sky
radiation
Check
every hour present,
values in range, dew
point ≤ air temperature,
solar parts add up
Flag
jumps where sources
meet: > 10 °C, 40 %
RH, 15 m/s, 10 hPa;
for review only
Shift
to local standard
time (UTC+1),
no summer time
Weather file + record file
EPW for one actual year; a JSON
record of settings, stations used,
test errors and share by source:
measured
own record
nearby
ERA5